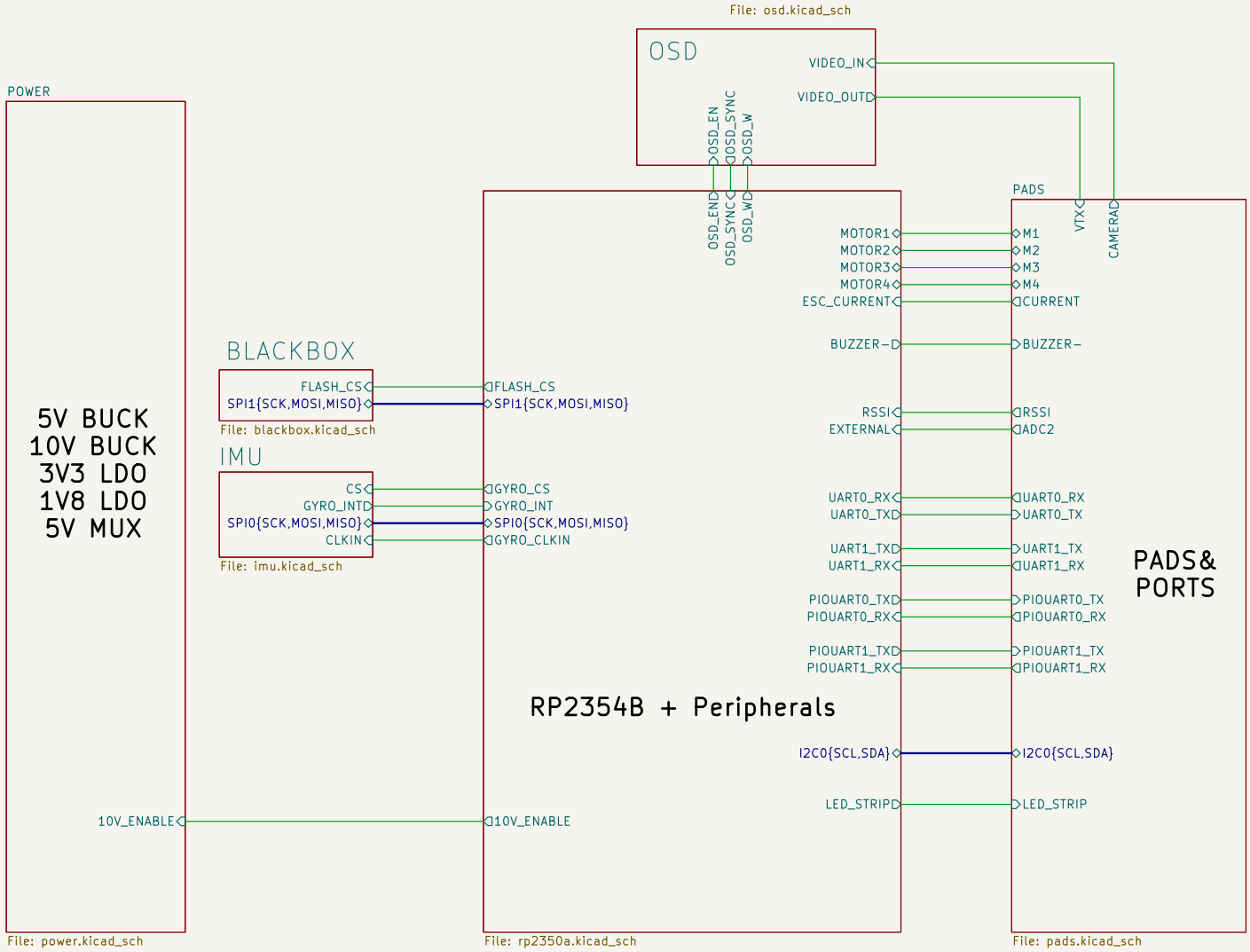


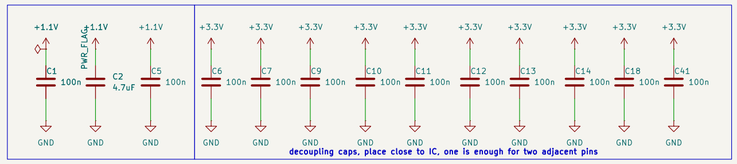
OpenDrone

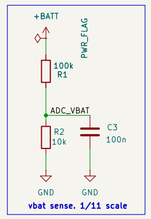
Sheet: /
File: OpenFC.kicad_sch

Title: Overview of OpenFC Lite (parent sheet)

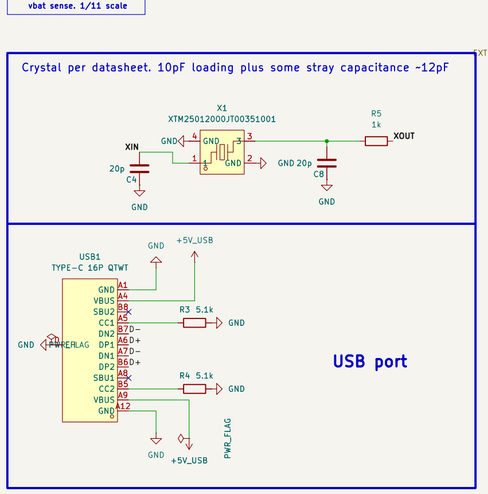
Size: A4	Date: 2026-11-12	Rev: 2-LW1
KiCad E.D.A. 10.0.8		Id: 1/7

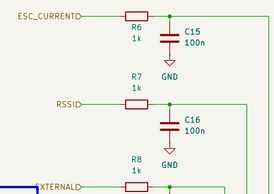
RP2354B: Larger variant with integrated flash (2MB) and more GPIO

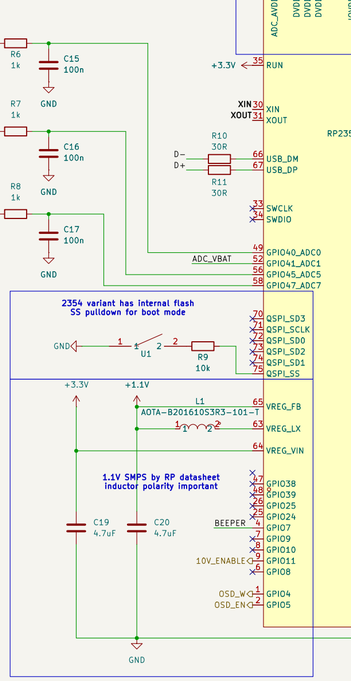


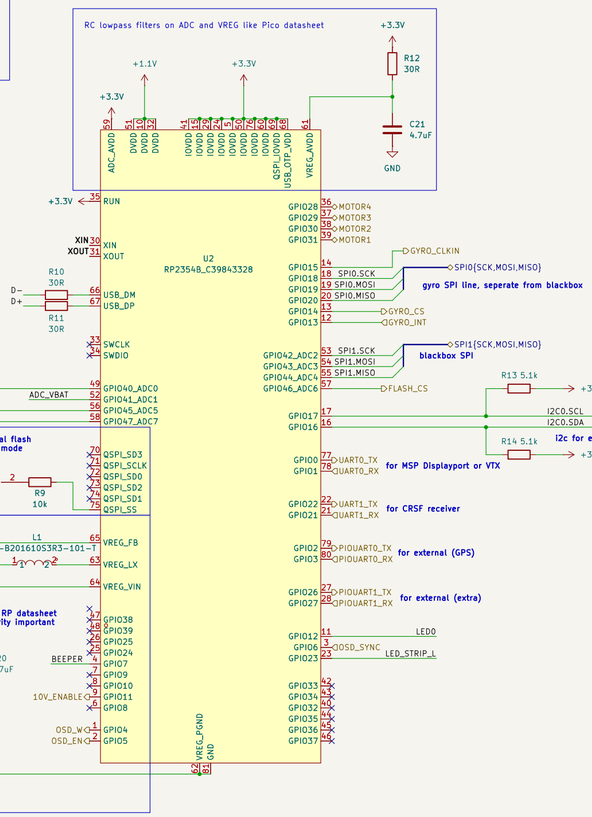


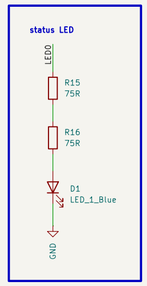
4.7u on opposite side of regulator

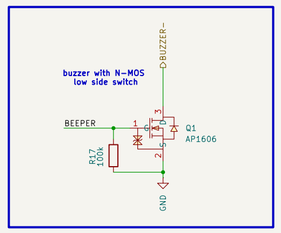


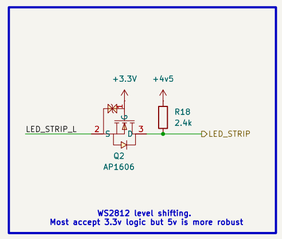












OpenDrone

Sheet: /RP2350A/
File: rp2350a.kicad_sch

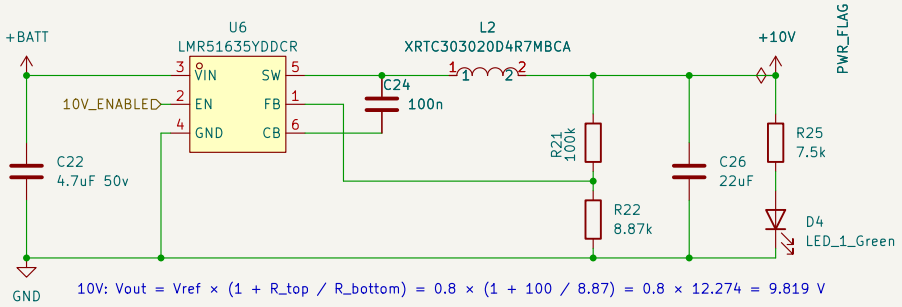
Title: RP2350 Pin mapping and universal peripherals

Size: A2	Date: 2026-11-12	Rev: LW1
KiCad E.D.A. 10.0.8		Id: 3/7

PFM version of the LMR51635 for better light-load efficiency and spread-spectrum EMI, at the cost of variable switching frequency at light load

10V (3.5A)

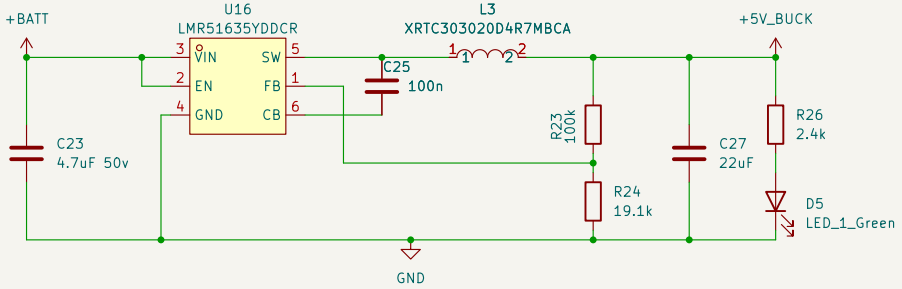
4.3 – 60V input
Synchronous buck
1.1MHz



10V: Vout = Vref × (1 + R_top / R_bottom) = 0.8 × (1 + 100 / 8.87) = 0.8 × 12.274 = 9.819 V

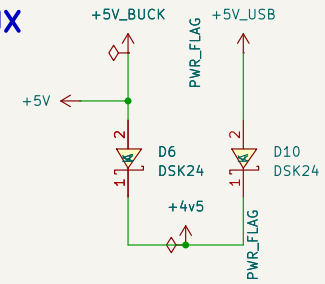
5V (3.5A)

4.3 – 60V input
Synchronous buck
1.1MHz

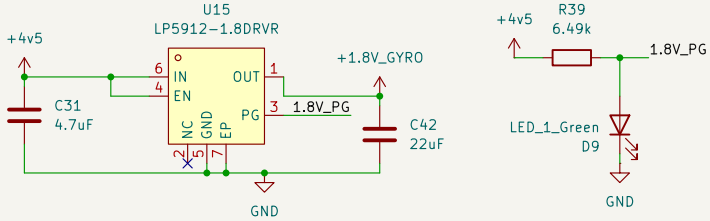


5V: Vout = Vref × (1 + R_top / R_bottom) = 0.8 × (1 + 100 / 19.1) = 0.8 × 6.236 = 4.988 V

5V Power MUX

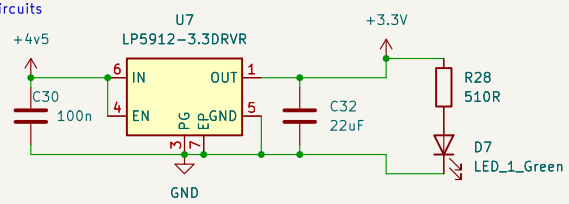


1.8V 500mA LDO (high PSRR)



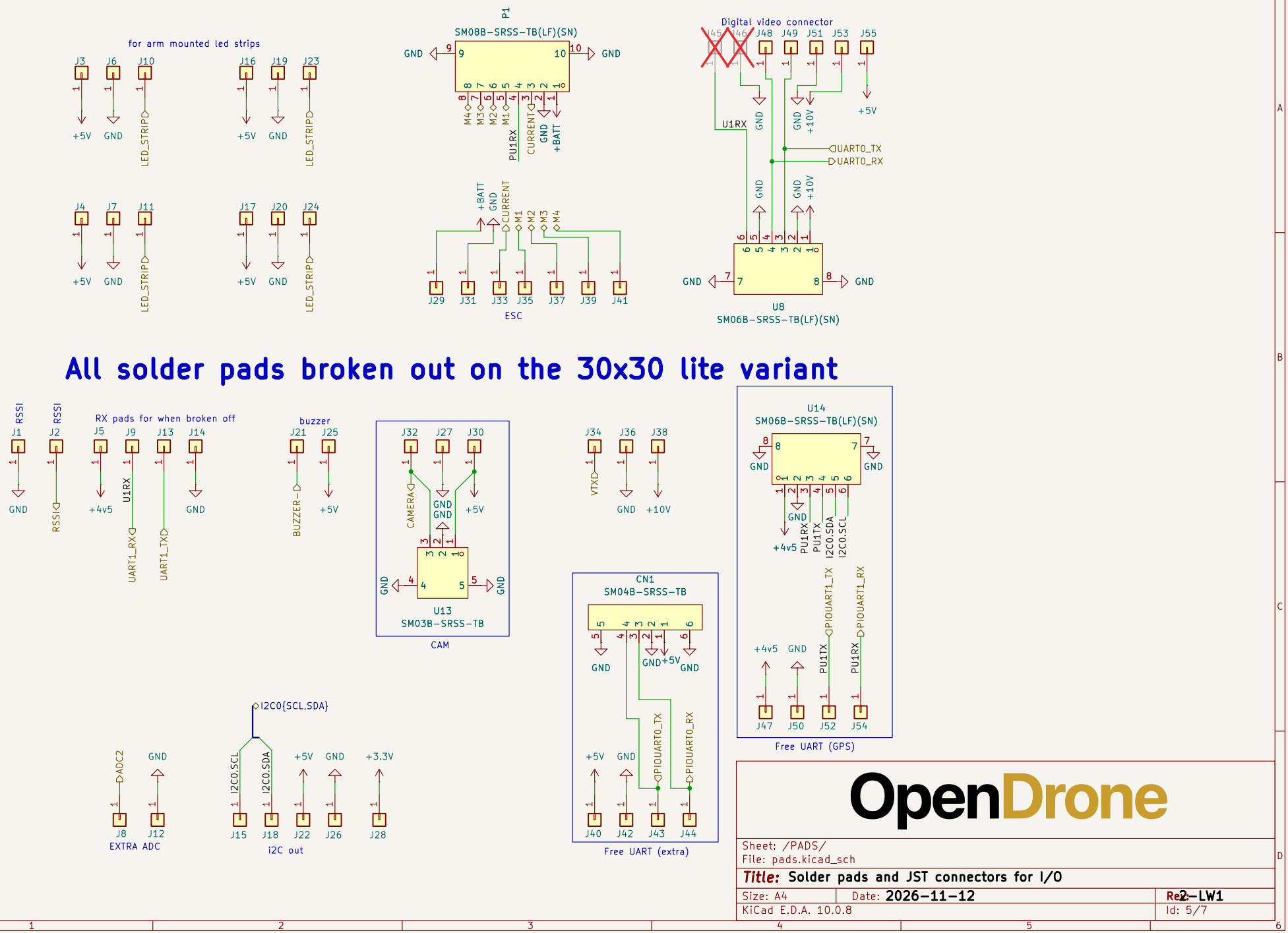
3.3V 0.5A MAIN LDO

Up to 6.5V input (5V)
Thermal Protection
Short circuit protection
Reverse current protection
Low noise for RF and analog circuits



OpenDrone

Sheet: /POWER/	
File: power.kicad_sch	
Title: Power delivery circuitry for flight controller	
Size: A4	Date: 2026-11-12
KiCad E.D.A. 10.0.8	Rev: 2-LW1
	Id: 4/7

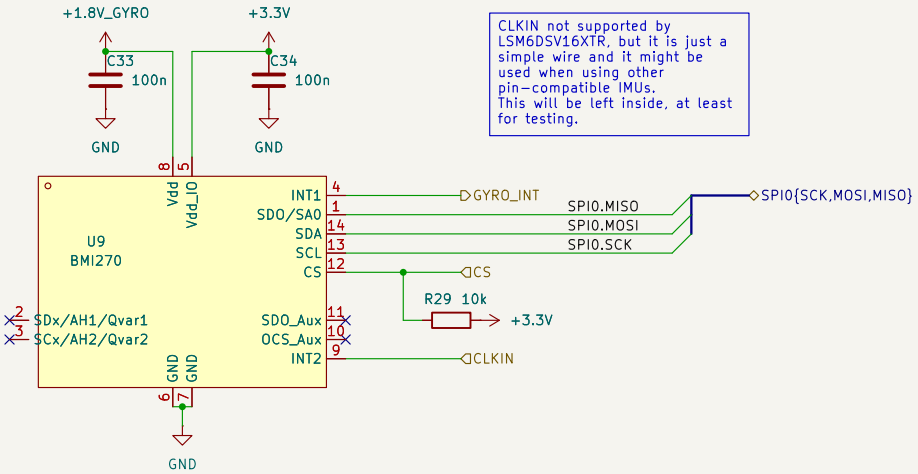


OpenDrone

Sheet: /PADS/	
File: pads.kicad_sch	
Title: Solder pads and JST connectors for I/O	
Size: A4	Date: 2026-11-12
KiCad E.D.A. 10.0.8	Id: 5/7

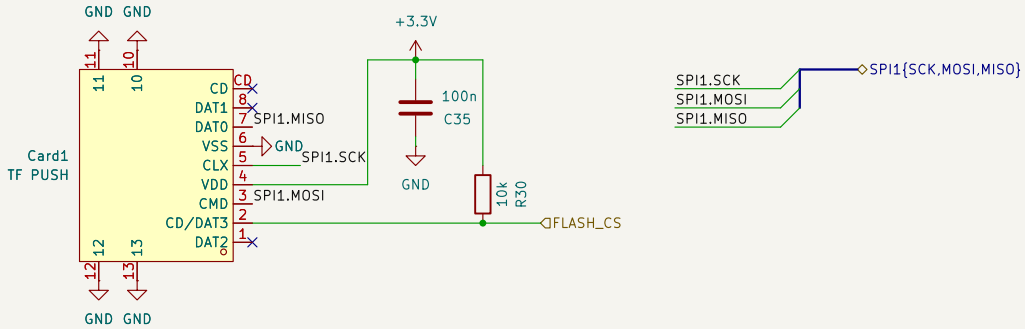
Rev: LW1

Universal IMU wiring and LGA footprint (BMI270, LSM6, TDK..)



OpenDrone

Sheet: /IMU/		
File: imu.kicad_sch		
Title: IMU SPI wiring and local decoupling		
Size: A5	Date: 2026-11-12	Rev:2-LW1
KiCad E.D.A. 10.0.8		Id: 6/7



OpenDrone

Sheet: /BLACKBOX/
File: blackbox.kicad_sch

Title: BLACKBOX SPI wiring and local decoupling

Size: A5	Date: 2026-11-12	Rev: LW1
KiCad E.D.A. 10.0.8		Id: 7/7

VIDEO PATH

Chroma Filter
fc = 500khz

DC Restoration

Sync detection, VID_DC clamped to ~-0.2V DC.

VIDEO_IN

75R input terminatiom

R31
75R

GND

R35
680R

TP2
TestPoint

VID_FILT

C36
470p

GND

C38
1uF

TP3
TestPoint

VID_DC

D8
SDM02U30LP3-7B

GND

+3.3V +3.3V

C40
100n

GND

U12

GND

IN-
IN+
VCC
VEE
OUT

TLV7031DPWR

OSD_SYNC

GND

Comparator detects sync pulses, read by RP2350 PIO

Resistor divider let's GPIO port select between
high (VOH = 1V: white) or low (VOL = 0.3v: black)

+3.3V

12k
R33

R32
3.9k

OSD_W

OSD_LVL

1.4k
R34

GND

	OSD_EN	OSD_W
black pixel	1	0
white pixel	1	1
transparant	0	x

+3.3V +3.3V

C37
100n

GND

U10
SN74LVC1G3157DTBR

B1
B2
A
S
VCC
GND

TP1
TestPoint

OSD_EN

GND

Analog switch (SPDT)

+3.3V +3.3V

C39
100n

GND

TP4
TestPoint

U11R38
COS8051SOT 75R

IN+
IN-
V+
V-
OUT

VIDEO_OUT

GND

R37
510R

R36
510R

GND

Non-Inverting Buffer output for low impedance.
6dB gain for 75R and 75R termination on VTX.

OpenDrone

Sheet: /OSD/
File: osd.kicad_sch

Title: Analog OSD using PIO State Machine on RP2350

Size: A3	Date: 2026-11-12	Rev: 2-LW1
KiCad E.D.A. 10.0.8		Id: 11/7